SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Machine Learning

ASSESSMENT 1 – Analytical Problem solving (5 QUESTIONS)

Course Code:	ITA05	**Course Title:**	**Computer Vision**
CO Assessed:	CO2	**Assessment:**	Assessment 1: Analytical Problem solving
Weightage:	35% of CIA	**Total Marks:**	(5 Questions × 10 Marks)
CO2 Statement:	***Apply image enhancement, filtering, and segmentation techniques to analyze images, detect edges and boundaries, and solve real-world Computer Vision problems.***		
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Section / Batch:	CSE	**Date:**	21/10/2026

Code of Conduct

I _____ (NAME/ REG NO) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

QUESTIONS: 5 **Total Marks: 50**

Annexure A

Analytical Problems

Q1. A grayscale image contains the following pixel intensity values: 10, 10, 20, 20, 20, 30, 30, and 40. Construct the histogram by calculating the frequency of each intensity value. Calculate the cumulative distribution function (CDF) for the given image. Apply histogram equalization using the formula $s = \text{round}((L - 1) \times CDF)$, where $L = 256$. Determine the new intensity values and compare the contrast distribution before and after equalization.

Q2. A 3×3 image has the intensity values [[10, 20, 30], [20, 100, 20], [10, 20, 30]]. Apply a 3×3 averaging filter to calculate the new intensity value of the centre pixel. Replace the centre pixel with the calculated average while keeping the other pixels unchanged. Apply a median filter to the same neighbourhood and calculate the centre pixel value. Compare the results of the averaging and median filters and determine which filter better preserves image details in the presence of noise.

Q3. Consider a grayscale image with the intensity matrix [[50, 50, 50], [50, 100, 50], [50, 50, 50]]. Apply the Laplacian filter mask [[0, −1, 0], [−1, 4, −1], [0, −1, 0]] to calculate the response at the centre pixel. Use the calculated response to sharpen the centre pixel by adding it to the original intensity value. Determine the resulting centre pixel intensity after sharpening. Explain how the Laplacian filter enhances intensity transitions and fine details in the image.

Q4. A grayscale image contains the pixel intensity values [[40, 60, 120], [80, 150, 200], [30, 100, 180]]. Apply global thresholding with a threshold value of 100, assigning 1 to pixels greater than or equal to the threshold and 0 to the remaining pixels. Construct the resulting binary image. Calculate the number of foreground and background pixels after thresholding. Explain how the selected threshold separates objects from the background and identify one limitation of this method.

Q5. A grayscale image has the intensity matrix [[10, 10, 10], [10, 50, 90], [10, 50, 90]]. Apply the Sobel operator using the masks Gx = [[−1, 0, 1], [−2, 0, 2], [−1, 0, 1]] and Gy = [[−1, −2, −1], [0, 0, 0], [1, 2, 1]] to calculate the gradient components at the centre pixel. Calculate the gradient magnitude using $G = \sqrt{(Gx^2 + Gy^2)}$. Determine the dominant edge direction based on the calculated gradient components. Explain how the detected edge information can be used for boundary detection and image segmentation.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Content Accuracy	30	Highly accurate, indepth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

ASSESSMENT 1 — SOLUTIONS

ITA05: Computer Vision | CO2 — Analytical Problem Solving

The following answers show the calculations and reasoning for each question. The original assessment pages are retained before this solution section.

Q1. Histogram, CDF and Histogram Equalization

Given: The grayscale pixel values are 10, 10, 20, 20, 20, 30, 30 and 40. The total number of pixels is N = 8, and the number of possible intensity levels is L = 256.

1. Construct the histogram

A histogram records how many pixels have each intensity. Counting the given values gives two pixels at intensity 10, three at intensity 20, two at intensity 30 and one at intensity 40. The frequencies add up to 8, which matches the total number of pixels.

Intensity	Frequency	Probability	Cumulative frequency	CDF
10	2	2/8 = 0.250	2	2/8 = 0.250
20	3	3/8 = 0.375	5	5/8 = 0.625
30	2	2/8 = 0.250	7	7/8 = 0.875
40	1	1/8 = 0.125	8	8/8 = 1.000

2. Calculate the cumulative distribution function (CDF)

The CDF at a particular intensity is the cumulative number of pixels up to that intensity divided by the total number of pixels. For example, the CDF at intensity 20 is (2 + 3)/8 = 5/8 = 0.625. The CDF reaches 1.0 at the highest observed intensity because all eight pixels have then been counted.

3. Apply histogram equalization

The question specifies s = round((L – 1) × CDF). Since L = 256, L – 1 = 255. Substituting each CDF value gives the following mapping:

```
For intensity 10: s = round(255 × 0.250) = round(63.75) = 64
For intensity 20: s = round(255 × 0.625) = round(159.375) = 159
For intensity 30: s = round(255 × 0.875) = round(223.125) = 223
For intensity 40: s = round(255 × 1.000) = 255
```

Therefore, the original sequence maps as follows:

10, 10, 20, 20, 20, 30, 30, 40 → 64, 64, 159, 159, 159, 223, 223, 255

Comparison: The original values lie between 10 and 40, so they occupy a relatively narrow part of the 0–255 intensity range. After equalization, the mapped values extend from 64 to 255. This spreads the observed levels across a wider range and can improve global contrast. The relative number of pixels in each group does not change; only the intensity assigned to each group changes.

Q2. Averaging Filter and Median Filter

Given neighbourhood:

```
[[10, 20, 30], [20, 100, 20], [10, 20, 30]]
```

1. Apply the 3 × 3 averaging filter

An averaging filter replaces the centre pixel with the arithmetic mean of all nine values in the 3 × 3 neighbourhood. First, add the values:

```
Sum = 10 + 20 + 30 + 20 + 100 + 20 + 10 + 20 + 30 = 260
Average = 260/9 = 28.888… ≈ 28.89
```

Thus, the new centre value is approximately 28.89, or 29 if an integer-valued pixel is required. All other pixels remain unchanged, as instructed.

10	20	30
20	28.89	20
10	20	30

2. Apply the median filter

The median filter sorts the same nine values from smallest to largest and selects the middle value. The sorted sequence is:

```
10, 10, 20, 20, 20, 20, 30, 30, 100
```

Because there are nine values, the fifth value is the median. Hence, the new centre pixel is 20.

10	20	30
20	20	20
10	20	30

Comparison and conclusion

The centre value 100 is much higher than the surrounding values. It raises the mean to approximately 28.89, whereas the median remains 20 because it is not strongly affected by a single extreme value. An averaging filter smooths an image but may blur edges and fine details. A median filter is generally more effective for removing impulse noise while preserving boundaries. For this neighbourhood, the median filter is the better choice.

Q3. Laplacian Filtering and Image Sharpening

Given image: [[50, 50, 50], [50, 100, 50], [50, 50, 50]]

Laplacian mask: [[0, −1, 0], [−1, 4, −1], [0, −1, 0]]

1. Calculate the Laplacian response

The mask is applied by multiplying each image value by the corresponding mask coefficient and adding the products. The centre coefficient is 4, the four direct neighbours have coefficient −1, and the corner coefficients are 0.

```
R = (4 × 100) − 50 − 50 − 50 − 50 = 400 − 200 = 200
```

The Laplacian response at the centre pixel is 200.

2. Sharpen the centre pixel

The question instructs us to add the response to the original centre value:

```
Sharpened value = 100 + 200 = 300
```

The calculated sharpened intensity is 300. Since an 8-bit grayscale image can store values only from 0 to 255, the value would be clipped to 255 when written to an 8-bit image. The unclipped mathematical result remains 300.

Explanation: The Laplacian measures rapid intensity changes using a second-order derivative. The centre pixel is 100, while each of its four direct neighbours is 50, so the mask produces a strong positive response. Adding this response as directed by the question increases the centre intensity and makes the transition between the centre and its surroundings more pronounced. In practical image sharpening, the sign of the Laplacian mask matters; the stated mask and the requested addition rule have been followed here.

Q4. Global Thresholding and Binary Segmentation

Given image: [[40, 60, 120], [80, 150, 200], [30, 100, 180]]

The threshold is T = 100. According to the question, each pixel with intensity greater than or equal to 100 is assigned 1; every pixel below 100 is assigned 0.

1. Apply the threshold to each row

```
Row 1: 40 → 0, 60 → 0, 120 → 1
Row 2: 80 → 0, 150 → 1, 200 → 1
Row 3: 30 → 0, 100 → 1, 180 → 1
```

2. Resulting binary image

0	0	1
0	1	1
0	1	1

Binary image = [[0, 0, 1], [0, 1, 1], [0, 1, 1]].

3. Count foreground and background pixels

Foreground (value 1): the pixels with original intensities 120, 150, 200, 100 and 180. Therefore, the foreground contains 5 pixels.

Background (value 0): the pixels with original intensities 40, 60, 80 and 30. Therefore, the background contains 4 pixels.

Explanation and limitation: Global thresholding separates pixels into two groups using a single threshold value. In this example, all pixels at or above 100 are classified as foreground. This method is simple and computationally inexpensive, so it is useful when object and background intensities are clearly different. Its main limitation is that one threshold may not work well when illumination varies across the image or when the foreground and background have overlapping intensity values. Such conditions can cause incorrect pixel classification.

Q5. Sobel Operator, Gradient Magnitude and Edge Direction

Given image: [[10, 10, 10], [10, 50, 90], [10, 50, 90]]

Sobel masks:

```
Gx = [[−1, 0, 1], [−2, 0, 2], [−1, 0, 1]]
Gy = [[−1, −2, −1], [0, 0, 0], [1, 2, 1]]
```

1. Calculate the horizontal gradient Gx

Multiply each image value by the corresponding coefficient in Gx, then add the products:

```
Gx = (−10 + 10) + (−20 + 180) + (−10 + 90) = 0 + 160 + 80 = 240
```

Therefore, Gx = 240.

2. Calculate the vertical gradient Gy

Using the Gy mask, the top row contributes −10 − 20 − 10 = −40, the middle row contributes 0, and the bottom row contributes 10 + 100 + 90 = 200.

```
Gy = −40 + 0 + 200 = 160
```

Therefore, Gy = 160.

3. Calculate the gradient magnitude

```
G = √(Gx² + Gy²) = √(240² + 160²)
= √(57,600 + 25,600) = √83,200 ≈ 288.44
```

The gradient magnitude at the centre is approximately 288.44.

4. Determine the dominant edge direction

The gradient angle is θ = tan■¹(Gy/Gx) = tan■¹(160/240) ≈ 33.69°. This is the direction of greatest intensity increase, measured from the positive horizontal axis. Because both components are positive, the gradient points down and to the right in the usual image-coordinate system. The edge itself is perpendicular to the gradient, so its orientation is approximately 123.69° (equivalently, −56.31°) modulo 180°.

Application: Sobel filtering estimates intensity changes in horizontal and vertical directions. A large gradient magnitude indicates a likely edge, while the gradient angle indicates the direction of the strongest change. By thresholding the gradient magnitude, likely edge pixels can be selected. These edges can help identify object boundaries and support segmentation. In practical applications, smoothing before edge detection and linking neighbouring edge pixels can reduce noise and improve boundary continuity.

CLASSROOM DISCUSSION & PRACTICAL DEMONSTRATION

ITA05 – Computer Vision | CO2 | Assessment 1

This assessment tool was discussed in class with the guidance and support of the faculty. The photographs below document the classroom session and the practical learning environment in which the assessment was reviewed.



Figure 1. Students working collaboratively during the classroom assessment discussion.

CLASSROOM EVIDENCE

Faculty-supported assessment session

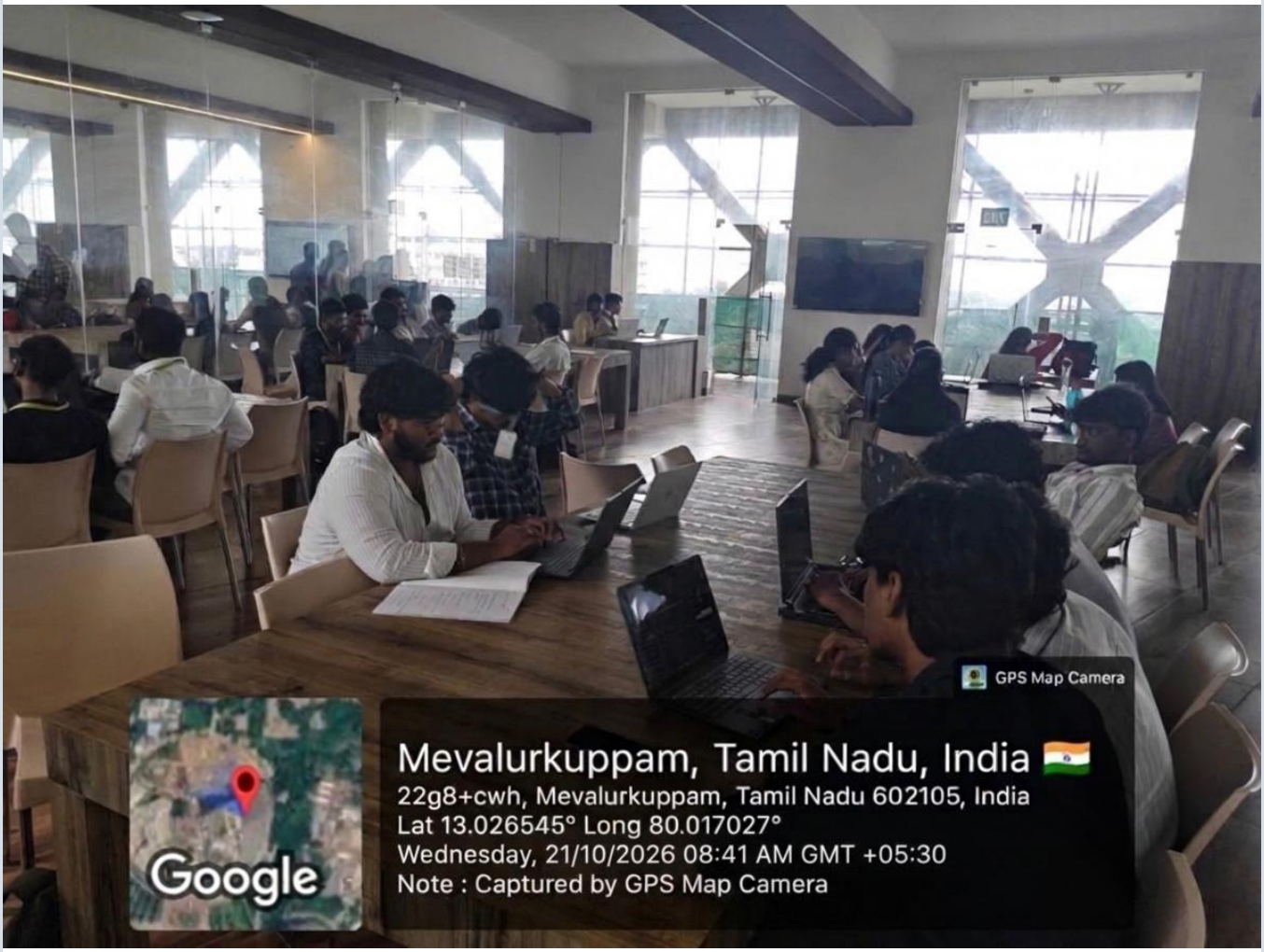


Figure 2. Classroom environment during the faculty-supported assessment session.